

Customer segments are clear. Forecast gains are not.

A reproducible eCommerce study of customer segmentation and observed-sales forecasting.

SEGMENTATION SAMPLE

6,000

customers before April 2018

FORECASTING SCOPE

5 products

selected using earlier sales

NOTEBOOK VALIDATION

36 cells

fresh-kernel execution passed

Tasks: unsupervised clustering + supervised regression. Repeat-purchase classification was retired.

Start with the correct unit of analysis

ORDERS

99,441

One order_id per order

CUSTOMERS

99,441

customer_unique_id links repeat buyers

ORDER ITEMS

112,650

Multiple item rows can belong to an order

Join on validated keys; remove invalid required values and duplicates. Count item rows for sales and distinct orders for customer frequency.

Approval timestamps define the event; approval does not prove delivery. Preserve valid sales history for modeling.

Build features before comparing models

CUSTOMERS

Recency, order frequency, recorded spending, product variety

Remove overlapping average spend and history length. Impute, log-transform skewed inputs, scale, and compare four features with 2D PCA.

PRODUCT SALES

Weekly: lags 1-4 + recent mean
Daily: calendar inputs only

Keep zero-sales periods. Use complete historical targets. Product IDs identify the daily models; they are not numeric predictors.

Price outlier filtering is for EDA only. It does not erase valid purchase history or target counts.

Respect time when tuning and evaluating

WEEKLY: 12 settings / 3 chronological validation windows

Training outcomes must finish before scoring. Final forecasts: 2 April-25 June 2018; targets end 22 July.

DAILY TRAIN

2017

2 Jan-31 Dec; select top five

DAILY VALIDATION

Jan-Mar 2018

Through 1 Apr; tune settings

DAILY TEST

Apr-Jul 2018

2 Apr-22 Jul; fixed models

These are historical backtests: evaluation dates were already viewed. Weekly target windows overlap.

K-Means gives a stable, broad two-group split

Common-sample silhouette: higher means clearer separation



Selected K-Means: 2 groups | 100% assigned | mean resampling ARI 1.00

Descriptive results for a 6,000-customer sample; strong separation does not prove marketing uplift.

Random Forest wins within ML, not overall



Weekly error: items per product-week, averaged across the four horizons. Lower is better.

RF error is 14.2% higher than the four-week mean. No forecasting improvement is claimed.

Calendar-only daily forecasts miss direction

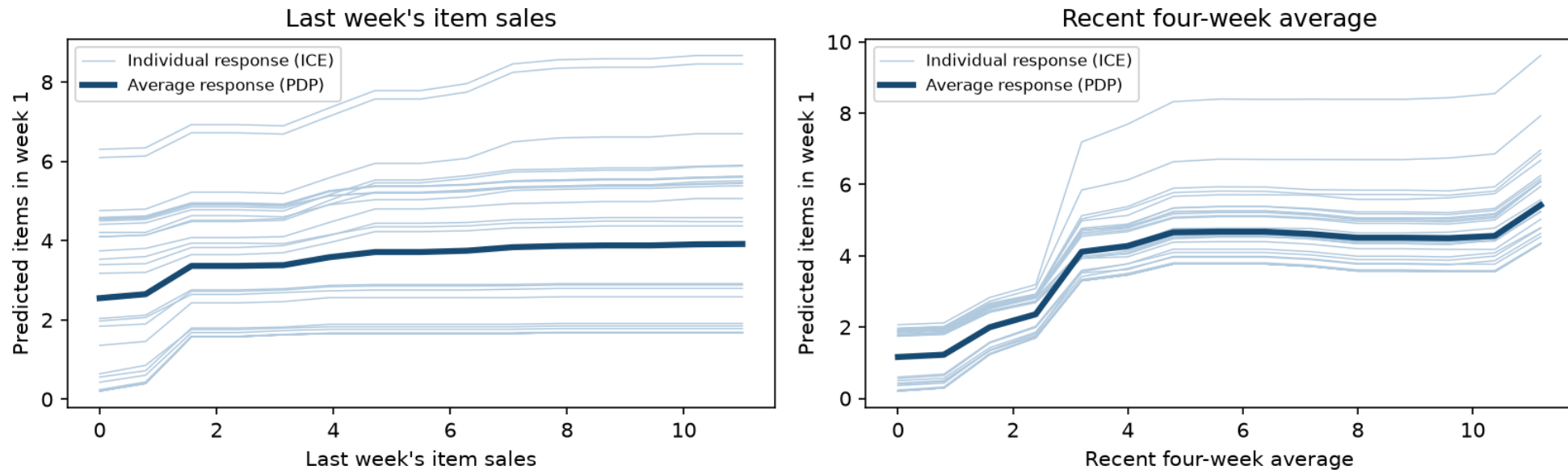
Daily product	Predicted trend	Observed trend	Daily error
1	Roughly flat	Upward	0.70 items
2	Roughly flat	Downward	1.27 items
3	Upward	Roughly flat	0.42 items
4	Roughly flat	Upward	0.96 items
5	Roughly flat	Downward	0.90 items

Trend: last 28 vs first 28 test days; changes within +/-0.1 items/day count as flat.

Trend matches: 0/5. Daily MAE: RF 0.850 vs average benchmark 0.817.

Recent activity drives the weekly predictions

How the weekly Random Forest responds to its inputs



Dark lines: average response (PDP). Pale lines: individual responses (ICE). Explanation is not causation.

Bias auditing exposes limits, not a fairness pass

STATE SELECTION GAP

3.95 pp

Hypothetical segment-only offer

MIN/MAX RATE RATIO

0.00

Driven by a small state sample

PROTECTED ATTRIBUTES

Missing

Gender, race, age and income

Both forecasting models underpredict positive-sales periods. Propose inclusive outreach, better coverage and human review; validate before deployment.

Geographic disparity does not establish discrimination. Sensitive-group fairness remains unmeasurable.

Reproducible evidence; clear remaining gaps

WHAT IS SAVED

- One executed notebook
- Fitted models + preprocessing
- Configs, seeds and source hashes
- Predictions, metrics and figures
- Targeted leakage/audit checks

WHAT COMES NEXT

- Validate on newly collected data
- Test practical mitigations
- Windows reproduction verified
- Combined final report complete
- Track results and limitations

Conclusion: segmentation is descriptive; neither forecasting RF has demonstrated an overall benchmark gain.